Expectation, Variance, MGF

1. Let a discrete random variable X have the following probability distribution:

X	1	2	3	4
$P(X)$	0.1	0.3	0.4	0.2

Find: (i) $E(X)$ (ii) $E(X^2)$ (iii) Var(X) (iv) $M_X(t)$

2. If X is a discrete random variable with $P(X = x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$ Find: (i) $E(X)$ (ii) $E(X^2)$ (iii) Var(X) (iv) $M_X(t)$.

3. Let X be a continuous random variable with PDF: $f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$.
Find (i) $E(X)$ (ii) $E(X^2)$ (iii) Var(X) (iv) $M_X(t)$.

4. Find the MGF, expectation, and variance of a random variable X with PDF: $f(x) = e^{-x}, \quad x \geq 0.$

5. Find the MGF, expectation, and variance of a discrete random variable with:

X	0	1
$P(X)$	0.6	0.4

6. If the MGF of a random variable is given by: $M_X(t) = e^{2t+3t^2}$ Find: $E(X)$, $Var(X)$.

7. Let X be a random variable with: $M_X(t) = \frac{1}{1-2t}, \quad t < \frac{1}{2}$. Find $E(X)$, $Var(X)$.

8. If $Y = 2X + 3$, where $E(X) = 4$ and $Var(X) = 5$, find: $E(Y)$, $Var(Y)$.

9. A fair die is rolled. Let X be the outcome. Find: $E(X)$, $Var(X)$.

10. A fair die is rolled twice. Let X be the sum of the outcomes. Find (i) $E(X)$ (ii) $E(X^2)$ (iii) Var(X) (iv) $M_X(t)$.

11. A fair coin is tossed three times. Let X be the number of heads obtained. Find (i) $E(X)$ (ii) $E(X^2)$ (iii) Var(X) (iv) $M_X(t)$.